Tisya Vaidya

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Education

Uppsala University — Uppsala, Sweden
M.S. in Data Science – Machine Learning and Statistics — *Aug. 2026 – Jun. 2028*

The Pennsylvania State University — University Park, PA, USA
B.S. in Computational Data Science, Minors in Mathematics and Statistics — *Aug. 2021 – May 2025*

Experience

Data Science Intern — Jun. 2026 – Aug. 2026
Digital Ipsum Pvt. Ltd. — *Remote*
- Developed a data-driven therapist–client matching system, designing the matching logic and integrating personalized recommendations into the company's website
- Built a machine learning model to analyze and predict digital marketing campaign ROI, applying data preprocessing, exploratory analysis, feature selection, and model evaluation

Applied Researcher – Autonomous Vehicles — Jan. 2025 – May 2025
PSU Autonomous Vehicle Team — *University Park, PA, USA*
- Researched stereo visual odometry techniques for GNSS-denied navigation
- Developed feature extraction, pose estimation, and trajectory reconstruction algorithms using OpenCV, Python, and ROS
- Implemented visual odometry pipelines with FLIR Blackfly S stereo cameras for depth mapping and trajectory reconstruction

Learning Assistant — Aug. 2023 – May 2025
The Pennsylvania State University — *University Park, PA, USA*
- Assisted in CMPSC 221: Object-Oriented Programming with Web-Based Applications, supporting students in understanding concepts
- Collaborated with the professor and TAs to refine course content, ensuring it met students' specific needs
- Graded exams, assignments, projects, and homework, ensuring accurate and fair evaluation of student performance

Facilitator — Aug. 2022 – May 2025
Women in Engineering Program — *University Park, PA, USA*
- Developed and delivered engaging lesson plans tailored to the CMPSC 131 Python curriculum
- Adapted assignments, exercises, and assessments to enhance student learning and exam readiness

Generative AI Intern — May 2024 – Jul. 2024
LTIMindtree — *Mumbai, India*
- **Image Generation:** Researched and implemented image generation techniques using models like DALL-E and Gemini, creating high-resolution 3D images
- **Instruction-Based Image Editing:** Analyzed research papers and contributed to developing and testing code for instruction-based image editing, enhancing image manipulation capabilities
- **Background Removal:** Implemented and tested background removal functionality using Microsoft Azure AI services
- **Web Scraping Automation:** Extracted eBay product details with Scrapy, BeautifulSoup, and Selenium, streamlining data collection

Data Science Intern — Jun. 2023 – Jul. 2023
Tata Consultancy Services — *Mumbai, India*
- Utilized BW/4HANA and Google Looker to convert raw client data in Excel into clean, accessible data, ensuring data integrity and accuracy
- Performed data analysis and cleansing to eliminate inconsistencies and errors, enhancing overall data quality
- Developed a front-end solution on the SAP AppGyver platform and integrated it with the back end

Projects

Enhancing Hate Speech Detection Through Multi-Model Ensembles | *Python, BERT, RoBERTa*
- Developed a hate speech detection system integrating transformer models (BERT and RoBERTa), ensemble techniques, and novel preprocessing on the Davidson 2017 dataset
- Designed a preprocessing pipeline with negation handling, lemmatization, and contextual text normalization
- Outperformed logistic regression and standalone transformers: 93% accuracy, macro F1 up from 0.70 to 0.79

LoanLens – Loan Approval Prediction System | *Python, Scikit-learn, Pandas, Matplotlib, Seaborn*
- Processed and analyzed 500+ loan applications with 12+ financial and demographic features, performing preprocessing, feature engineering, and EDA
- Implemented and evaluated Logistic Regression, KNN, SVM, Decision Tree, and ensemble methods, achieving 81% accuracy with ensembles

MindScope – Workplace Mental Health Analytics Dashboard | *Tableau*
- Created an interactive Tableau dashboard analyzing 1,200+ survey responses, using filters and calculated fields to explore links between demographics, workplace policies, and outcomes

Technical Skills

Programming & Libraries: Python, SQL, NumPy, Pandas, Scikit-learn, Matplotlib
Data & ML: Machine Learning, Data Analysis, Tableau, Power BI, Microsoft Excel
Languages: English, Hindi, Marathi, Swedish (A2)